

Real-Time SAR Ship Detection: A Speckle-Model-Based DSP Pipeline

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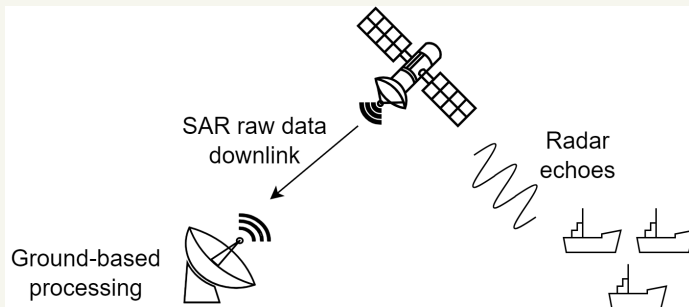
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Outline

- ▶ Introduction
- ▶ Problem and Motivation
- ▶ Highlight of Contribution
- ▶ System Definition
- ▶ Previous Methods
- ▶ Proposed Method
- ▶ Comparison
- ▶ Conclusion

Introduction

- ▶ SAR raw data are downlinked for ground-based processing, including SAR focusing, image enhancement, and ship detection.
- ▶ Detected ships provide information for maritime surveillance.

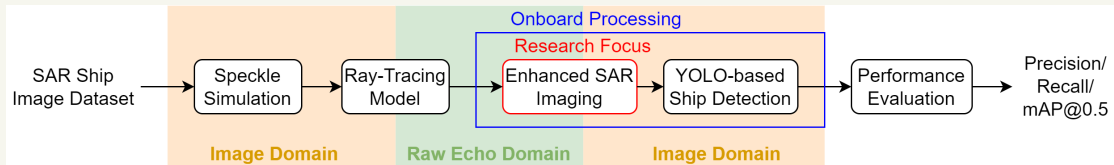


Problem and Motivation

- ▶ Ground-Based Processing
 - ▶ Large raw-data volume
 - ▶ Limited downlink bandwidth
 - ▶ Delayed target information
- ▶ Onboard Processing
 - ▶ Reduced data transmission and detection latency
 - ▶ Challenge: Limited computation, memory, and power
- ▶ Research Goal
 - ▶ Lightweight SAR processing for low-latency ship detection

Evaluation Framework

- Research Focus: Enhanced SAR imaging
- Evaluation Task: YOLO-based ship detection



► Evaluation Metrics

- Precision: $\frac{TP}{TP+FP}$
- Recall: $\frac{TP}{TP+FN}$
- mAP@0.5: Mean AP across classes at IoU threshold = 0.5

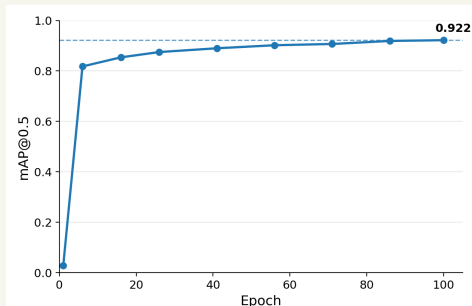
TP: True Positive, FP: False Positive, FN: False Negative

Dataset and Detection Model Setup

- ▶ HRSID
 - ▶ High-Resolution SAR Images Dataset
 - ▶ Training: 3,642 images
 - ▶ Validation: 1,962 images
 - ▶ Image size: 800×800 pixels
- ▶ YOLO Detection Model
 - ▶ YOLO11m fine-tuned on HRSID (100 epochs)

Validation Performance

- ▶ Precision: 0.910
- ▶ Recall: 0.840
- ▶ mAP@0.5: 0.922



YOLO Inference Result

► Hardware: Intel Core i9-12900K

	Inshore + Offshore (1962 images)	Inshore (369 images)	Offshore (1593 images)
Precision	0.913	0.8034	0.9818
Recall	0.846	0.7406	0.9651
mAP@0.5	0.918	0.8170	0.9838